SHRI RAMSWAROOP MEMORIAL UNIVERSITY



SESSION: 2026-27

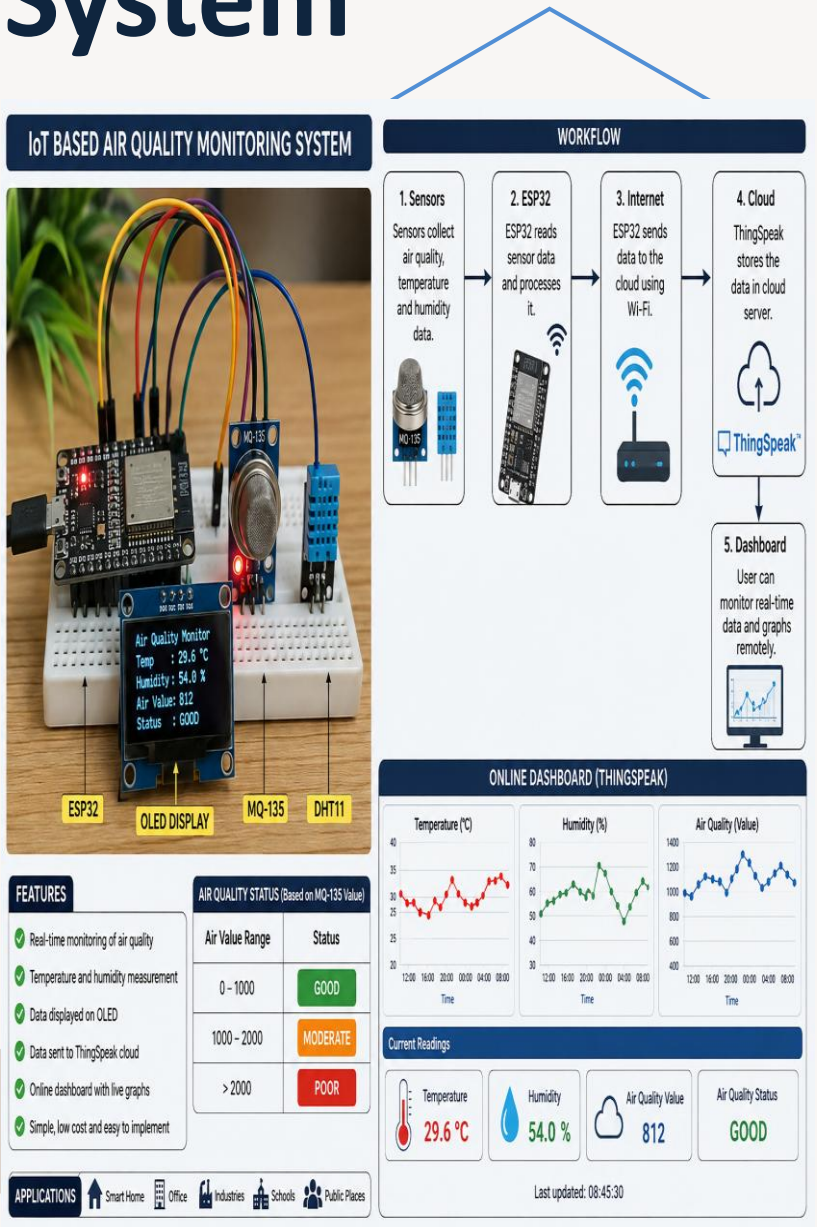
PROJECT:IOT BASED AIR &TEMP MONITORING SYSTEM

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IoT-Based Air Quality Monitoring System

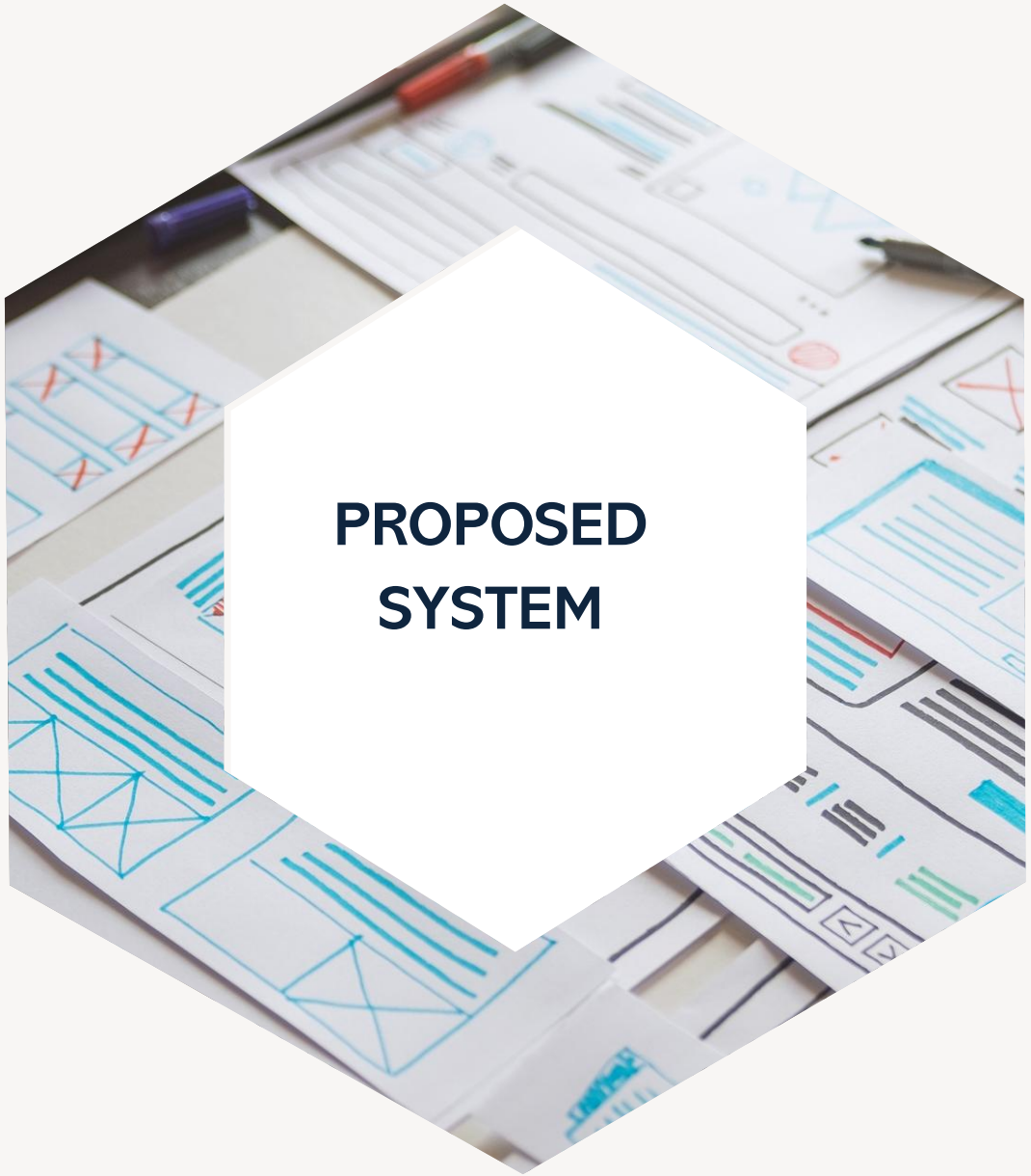
- Introduction:
- Air quality is an important factor affecting human health and environmental conditions. Air pollution can be caused by vehicle emissions, industrial activities, smoke, dust, and harmful gases.
- The Internet of Things (IoT) allows physical devices to collect and exchange data through the internet. By combining sensors, a microcontroller, and cloud technology, air-quality-related readings can be monitored locally and remotely.
- Our project develops a low-cost system that measures air-quality sensor readings, temperature, and humidity and displays the information on an OLED screen and an online dashboard.



Project Objectives

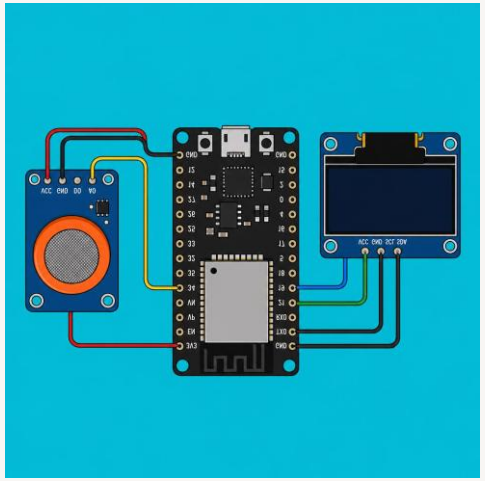
- The main objectives of this project are:
- To develop a low-cost IoT-based air monitoring prototype.
- To measure environmental temperature and relative humidity
- To collect air-quality-related readings using the MQ-135 sensor.
- To process sensor readings using the ESP32 microcontroller.
- To display readings on an OLED screen.
- To transmit sensor data over Wi-Fi.
- To visualize historical readings through the ThingSpeak cloud platform.



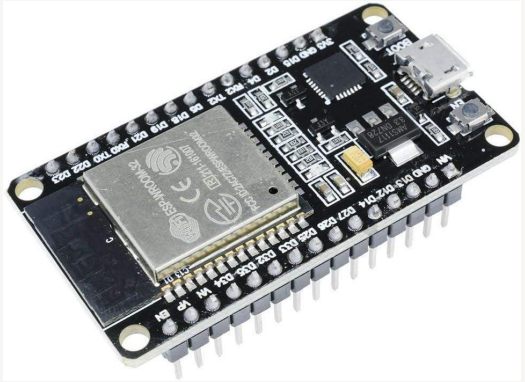


PROPOSED SYSTEM

- The proposed system consists of sensors connected to an ESP32 microcontroller. The sensors collect environmental data, and the ESP32 reads and processes these measurements.
- The results are displayed on a small OLED screen. Using built-in Wi-Fi, the ESP32 sends the collected data to ThingSpeak, where it can be stored and visualized through graphs.
- **Main functions:**
- Environmental data collection
- Local data display
- Wireless data transmission
- Online monitoring and visualization

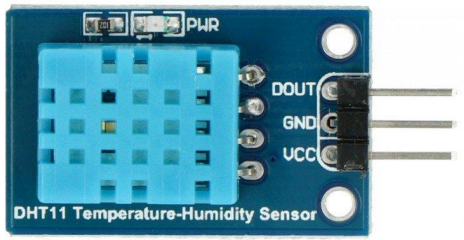


Hardware Components Required



1. **ESP32 DevKit V1** , Processes sensor data and provides Wi-Fi connectivity

2.**MQ-135 sensor** . Provides air-quality-related gas sensor readings



3.**DHT11 sensor** , Measures temperature and relative humidity

4. **0.96-inch OLED display** ,Displays readings locally

5. **Breadboard** , Supports circuit assembly

6. **Jumper wires** ,Connect the electronic components

7. **USB cable** and power source , Powers and programs the ESP32

8. 10 kΩ and 20 kΩ resistors ,Form a voltage divider to protect the ESP32 analog input when required

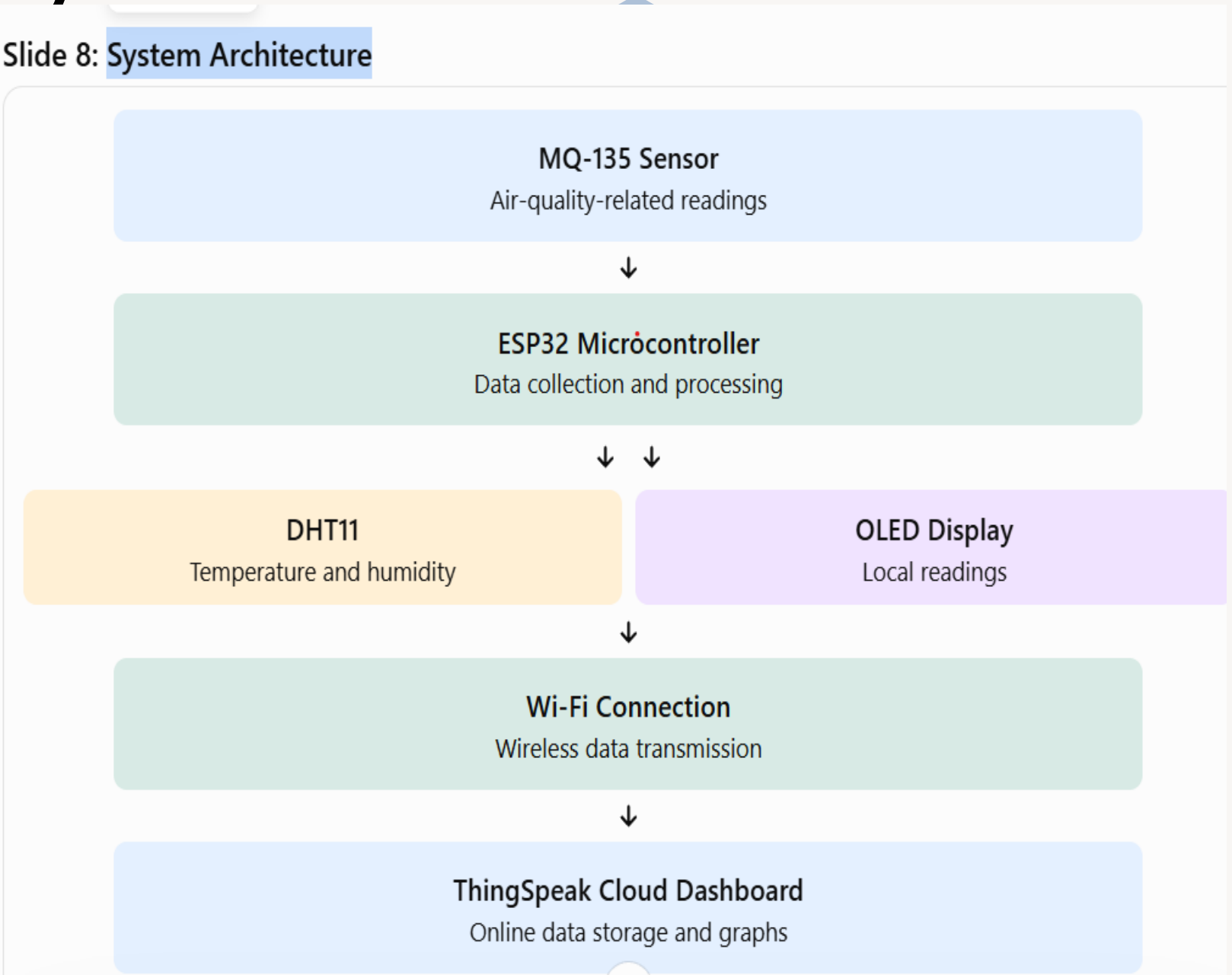


Software and Technologies Used

- Arduino IDE: Used to write, compile, and upload the program.
- Embedded C/C++: Used to program the ESP32.
- ESP32 Wi-Fi: Provides wireless network connectivity.
- ThingSpeak: Receives, stores, and visualizes sensor data online.
- DHT sensor library: Reads temperature and humidity.
- Adafruit GFX and SSD1306 libraries: Control the OLED display.
- ThingSpeak library: Helps transmit sensor readings to the cloud.

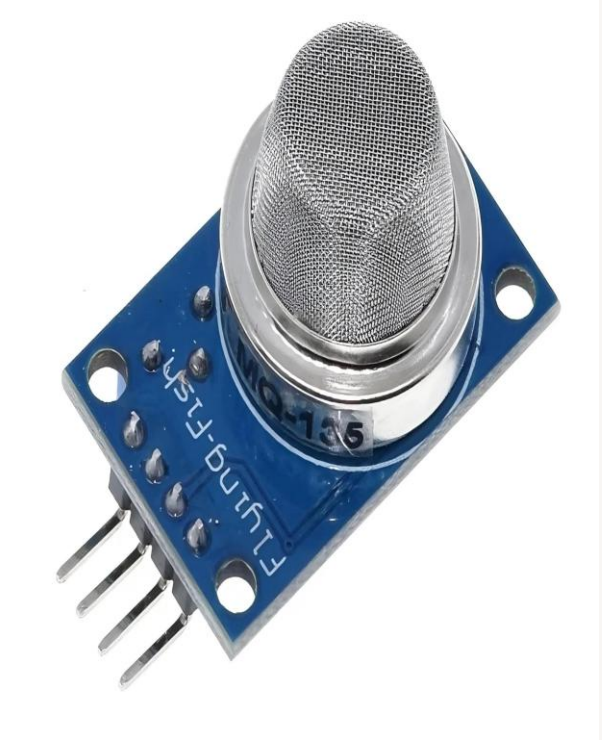
System Architecture

Slide 8: System Architecture



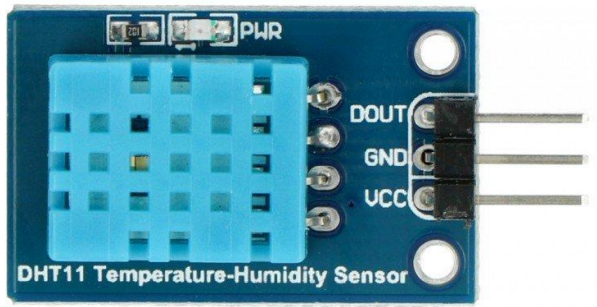
Working of the MQ-135 Sensor

- The MQ-135 is a gas-sensing module commonly used in air-quality-related demonstration projects.
- Working principle:
- The sensing element responds to certain gases in the surrounding environment.
- The module produces an electrical signal that changes with the sensing conditions.
- The ESP32 reads the analog output through its ADC input.
- The sensor value can be monitored over time to observe changes in the environment.
- Important limitation: The raw MQ-135 reading is not a direct AQI measurement. Accurate pollutant concentration or official AQI values require suitable calibration and appropriate measurement methods.



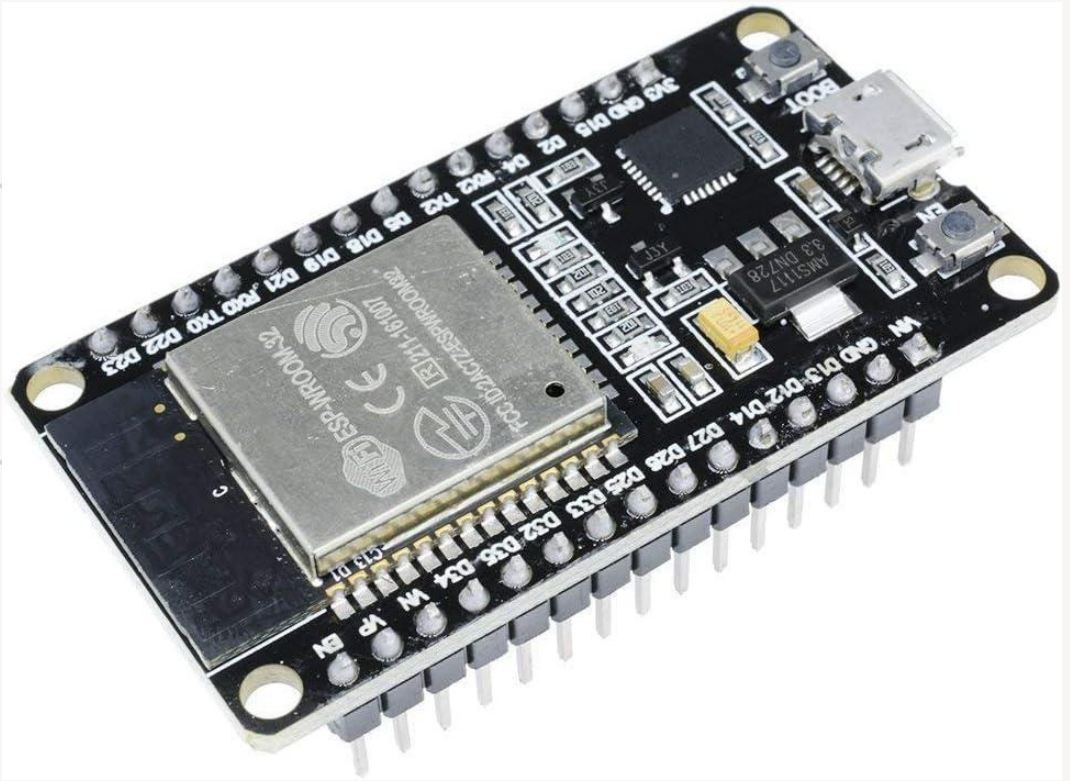
Temperature and Humidity Measurement

- The DHT11 sensor measures two important environmental parameters.
- Temperature: Measures the surrounding air temperature in degrees Celsius (°C).
- Relative humidity: Measures the amount of moisture in the air as a percentage (%).
- Digital output: Sends measurement data to the ESP32 through a data pin.
- Environmental context: Helps observe how temperature and humidity vary alongside air sensor readings.
- The DHT11 is affordable and easy to use, making it suitable for a basic college prototype. Its accuracy is limited compared with more advanced sensors.

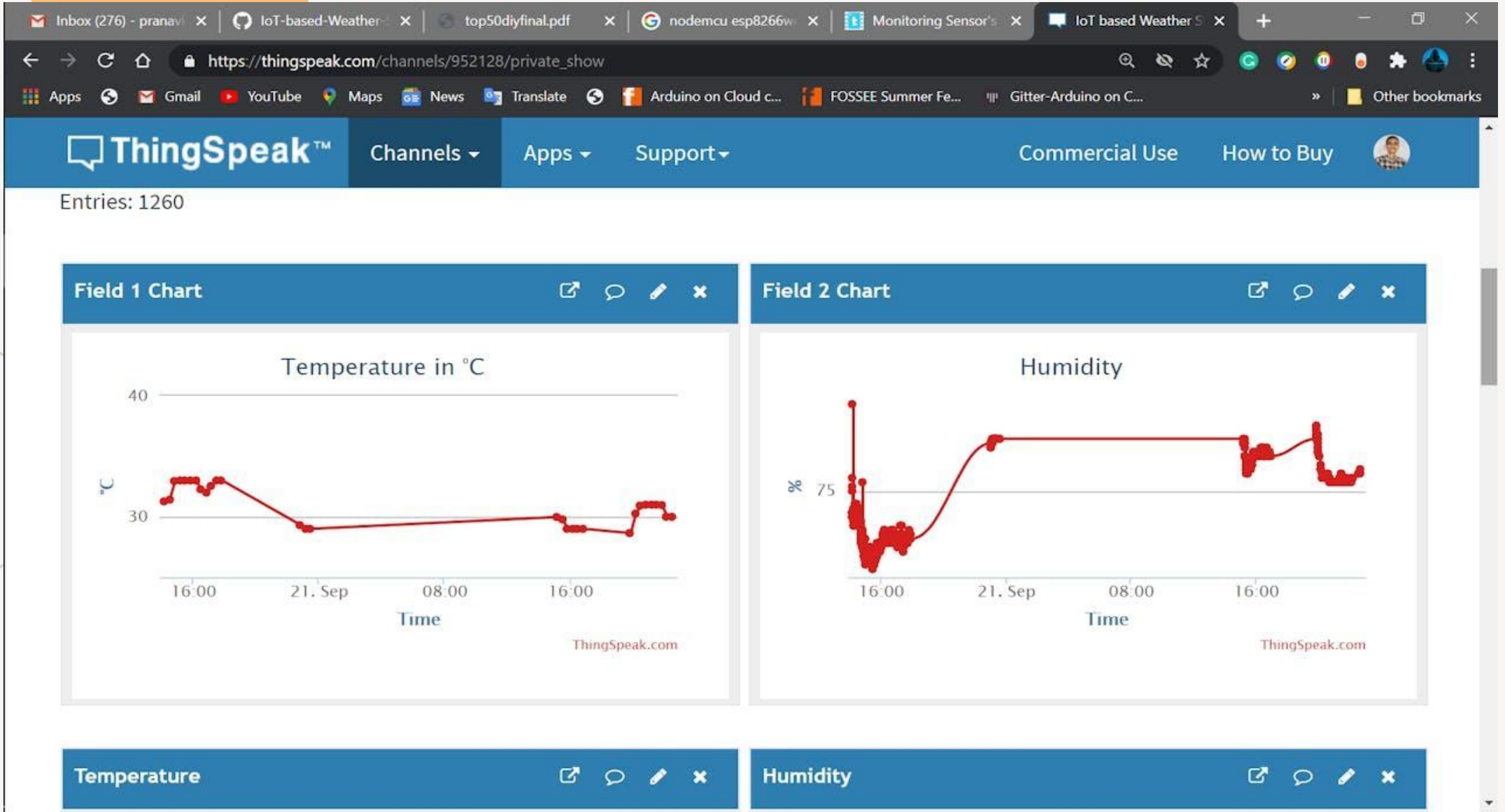


ESP32 Microcontroller

- The ESP32 acts as the main controller of the project.
- Key responsibilities:
- Reads the MQ-135 analog signal.
- Receives temperature and humidity data from the DHT11.
- Processes and organizes the sensor readings.
- Updates the OLED display.
- Connects to a wireless network using built-in Wi-Fi.
- Transmits readings to the ThingSpeak cloud platform.
- Its built-in wireless connectivity and analog input capability make it suitable for low-cost IoT applications.



IoT Cloud Integration — ThingSpeak



System Workflow and Expected Output

1. Sensors collect environmental data

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2. ESP32 reads and processes measurements

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3. OLED displays local readings

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4. ESP32 sends data through Wi-Fi

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5. ThingSpeak receives and stores data

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6. User views readings and graphs online

Illustrative OLED output:

Plain text

```
Air Quality Monitor
Temp: 29 C
Humidity: 54%
Air: 850
Status: GOOD*
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Conclusion and References

- The IoT-Based Air Quality Monitoring System demonstrates how sensors, microcontrollers, wireless communication, and cloud computing can be combined in a single prototype.
- The ESP32 collects sensor data, the OLED displays readings locally, and ThingSpeak enables online visualization. The project provides a practical introduction to IoT system design, environmental sensing, and remote data monitoring.
- Further development with calibrated, higher-accuracy sensors can improve measurement reliability and expand its practical applications.

References

- ESP32 technical documentation — Espressif Systems.
- MQ-135 sensor module datasheet.
- DHT11 sensor datasheet.
- Arduino IDE documentation.
- Adafruit SSD1306 and GFX library documentation.
- ThingSpeak IoT platform documentation.